

Registration Number: _____
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 Branch & Section: _____

SRM UNIVERSITY – AP, ANDHRA PRADESH

End Term Examination, May 2025

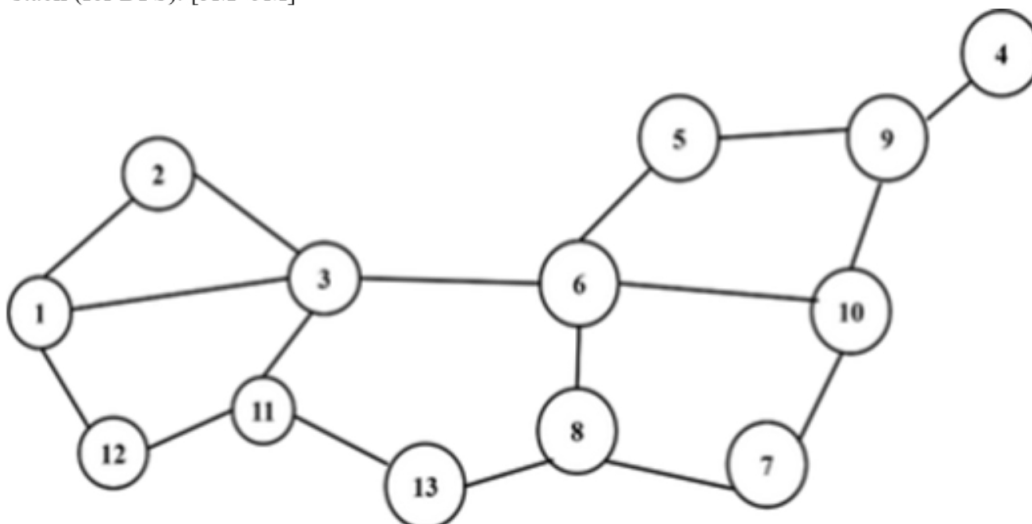
[Question Paper ID: 008959]

Subject	: DATA STRUCTURES	Subject	: CSE 102
Title		Code	
Batch	: 2024	Max Marks	: 50
Degree	: B.Tech., M.Tech. Integrated	Duration	: 2 hours
Branch	: CSE	QP Set	: ---

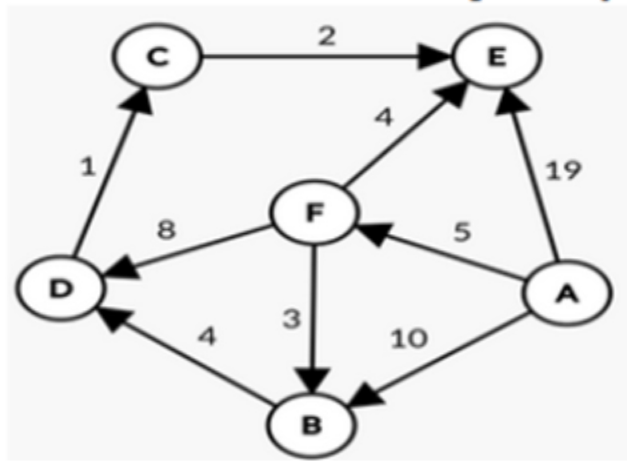
PART-A (5 × 10 Marks = 50 Marks)

Answer **any 5** Questions

- | | | | Marks | BL | CO |
|----|---|----|-------|----|-----|
| 1. | Develop pseudocode to implement a circular queue using arrays and perform the operations for Enqueue, Dequeue, and Display, with proper handling for overflow and underflow conditions. [10M] | 10 | 10 | 1 | 1,2 |
| 2. | a. Write an algorithm to check whether a doubly linked list is palindrome or not . [5M]
b. Write an algorithm to count the number of nodes in a doubly linked list. [5M] | 10 | 10 | 3 | 2,3 |
| 3. | What is a Binary Search Tree (BST)? Make a BST for the following sequence of numbers: 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48 . Traverse the tree in Pre-order, In-order and Post-order. [1M+6M+1M+1M+1M] | 10 | 10 | 2 | 1,2 |
| 4. | Give an example of a situation where an AVL tree would perform better than a regular BST. Construct AVL tree for the following data: 21,26,30,9,4,14,28,18,15,10,2,3,7 . Delete the node 28 and reconstruct the AVL tree. Compare the time complexities of AVL tree operations (insertion, deletion, search) with those of an unbalanced BST. [2M+4M+2M+2M] | 10 | 10 | 3 | 2,3 |
| 5. | Write pseudocode for the Breadth First Search (BFS) and Depth First Search (DFS) algorithms. Then, using the graph shown in the figure below, perform BFS and DFS traversals and print the traversal steps. Explain each step in detail by illustrating the contents of the queue (for BFS) and stack (for DFS). [5M+5M] | 10 | 10 | 3 | 2,3 |



6. a. Write Dijkstra's algorithm and provide a brief explanation of how it works. [5M]
 b. SRM University-AP has launched an emergency medical drone delivery system connecting various health centers in the city. The delivery points are represented as nodes (A to F), connected via one-way drone corridors with each corridor labelled with the travel time in minutes. The central dispatch hub is located at Node A, and the drone always starts its delivery from Node A. For each emergency request, the drone must reach a particular Centre in the shortest possible time. Using Dijkstra's algorithm, find the shortest time from Node A to all remaining nodes. [5M]



10 10 4 2,3

7. a. Write the pseudocode for the Quick Sort algorithm. Explain how the pivot is chosen and how the array is divided around the pivot during each recursive call. What is the time complexity in the best, worst, and average case scenarios. [5M]
- b. Apply the Quick Sort algorithm to sort the following array: $A = [9, 3, 7, 1, 6]$. Show all intermediate steps, including: [5M]
- Pivot selection (assume the last element is always chosen as the pivot),
 - Result after each partitioning,
 - Final sorted array.

10 10 3 2,3

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